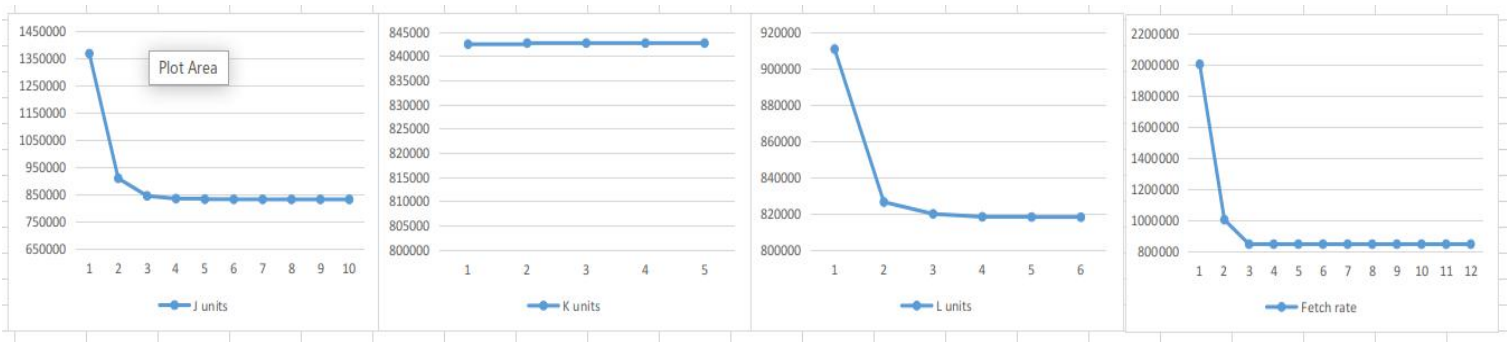
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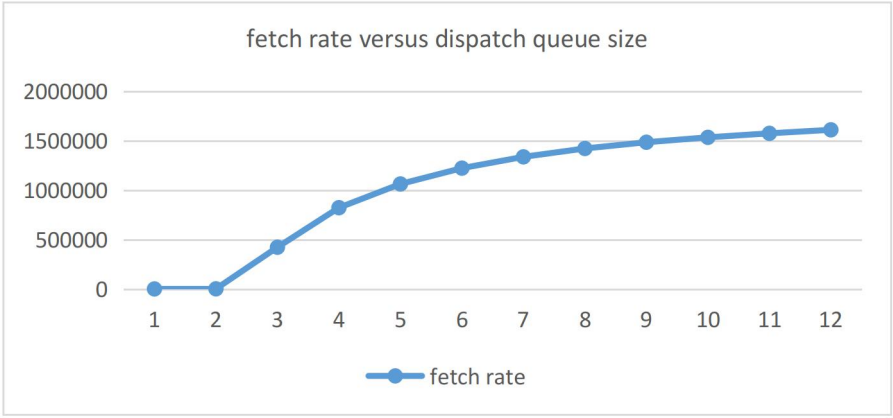
Part 1

Before performing the experiments, we should first determine the ranges of parameters. For fetch rate, it won't make sense for F to exceed the ROB size 12, which is also the effective size of scheduling queue. Therefore, the maximum number of other function units also should not exceed 12, because there can be at most 12 same-category function units being used at the same time. We will first study the effects of varying those parameters.



The above figures are plotted using the number of function units versus final cycle counts running bwaves603_2M.trace. By keeping other parameters default and varying J, K, L, F, we conclude that in multiple testcases, increasing J above 8 or 9, increasing K units above 1 or 2, and increasing L units above 5 or 6, and increasing fetch rate above 3, do not bring better performance. These results are expected because the trace decides how many function units it can utilize at max. Increasing function units more than that value will cause underutilization of function units.

With regard to the dispatch queue size, we found that in multiple scenarios, F=3 keeps the dispatch queue small while ensuring high throughput. A lower fetch rate does not fully utilize the function units, and a higher fetch rate makes instructions pile up in the dispatch queue.



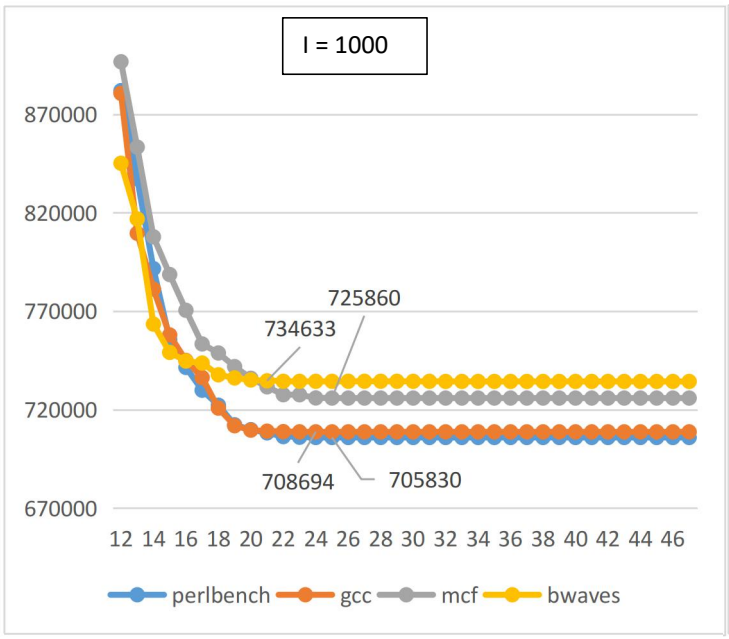
Finally, running a small set of parameters on every trace: F = [3, 4], J = [7, 9], K = [1, 2], L = [5, 7], we get the results of optimal function units sizes, as shown in the following chart.

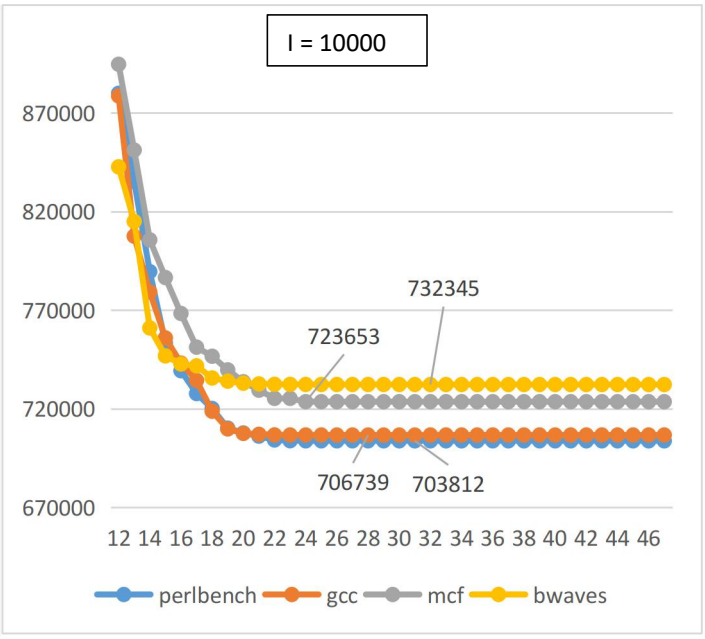
Parameter \ trace	bwaves	gcc	mcf	perlbench
F	3	3	3	3
J	8	8	8	9
K	2	1	1	1
L	6	6	5	6
Max DispQ Size	393908	429015	439301	449591
Final Cycle Count	829437	864715	855036	858731

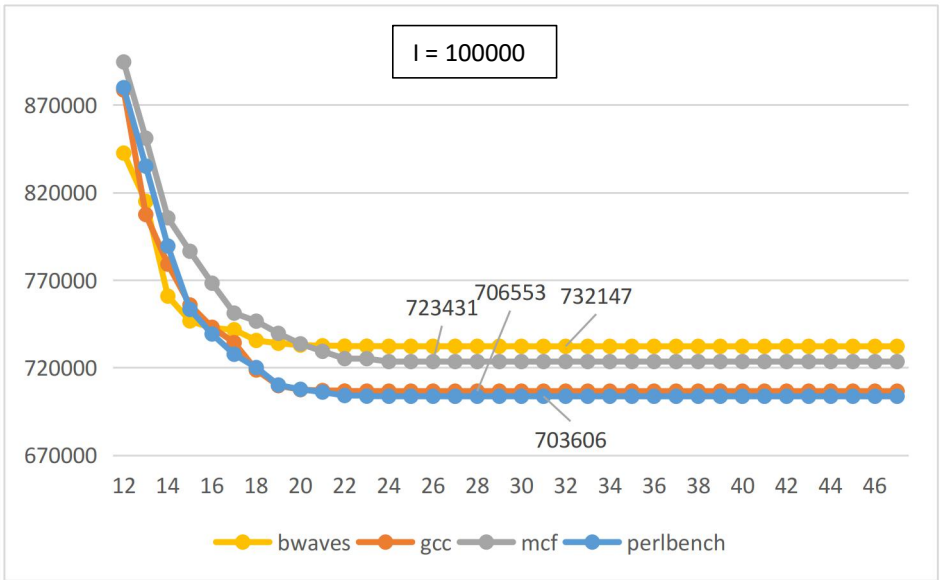
Part 2

Before experiments, I expect that to see the trend that the maximum cycles count goes down with increasing ROB size but stops decreasing at some point. This is because that a bigger ROB size help better utilizing the function units and improve instruction level parallelism. However, when ROB gets too big, the performance will be, in turn, limited by the number of function units. At this point, a bigger ROB does not bring any performance improvements.

This part is relatively straightforward. We will just start increasing the size of ROB from 12. The resulting graphs and optimal ROB sizes are shown below. Note the y-axis is the final cycle count for each trace.







Optimal ROB sizes for I values and traces:

Interrupt \ trace	bwaves	gcc	mcf	perlbench
I = 1000	32	24	25	25
I = 10000	32	28	24	31
I = 100000	32	28	26	31